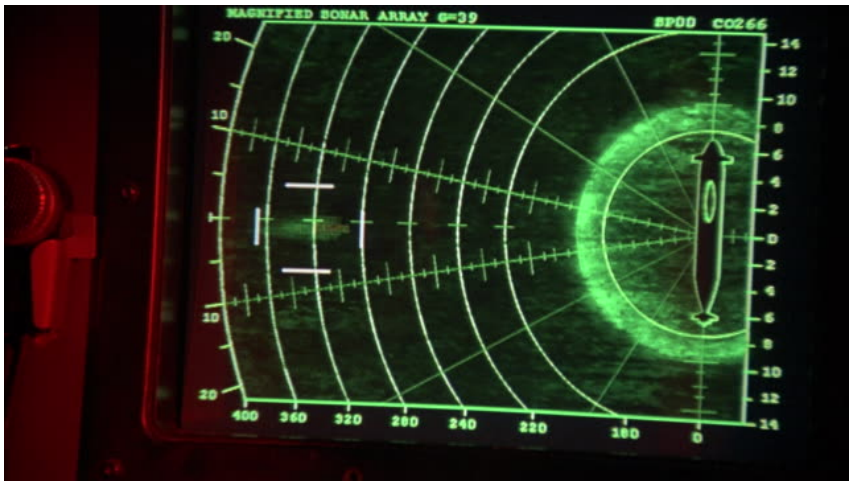
Two-Choice Signal Detection Model

Michael D. Lee

Signal Detection Theory

- Signal detection theory (SDT) is widely used in science and engineering for analyzing errors in two-choice decisions in which a decision process is trying to detect a "signal" instead of "noise" (Green & Swets, 1966; Macmillan, 2001)
  - e.g., sonar operators on a submarine trying to detect missiles
  - e.g., art evaluators trying to detect forgeries

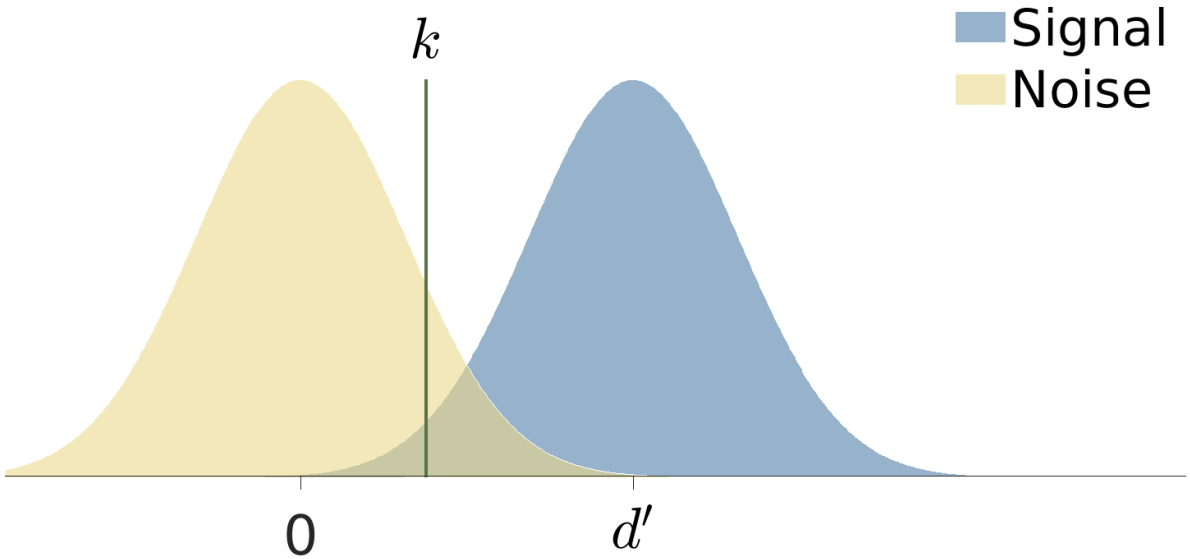
	really signal	really noise
detect signal	hit	false alarm
detect no signal	miss	correct rejection





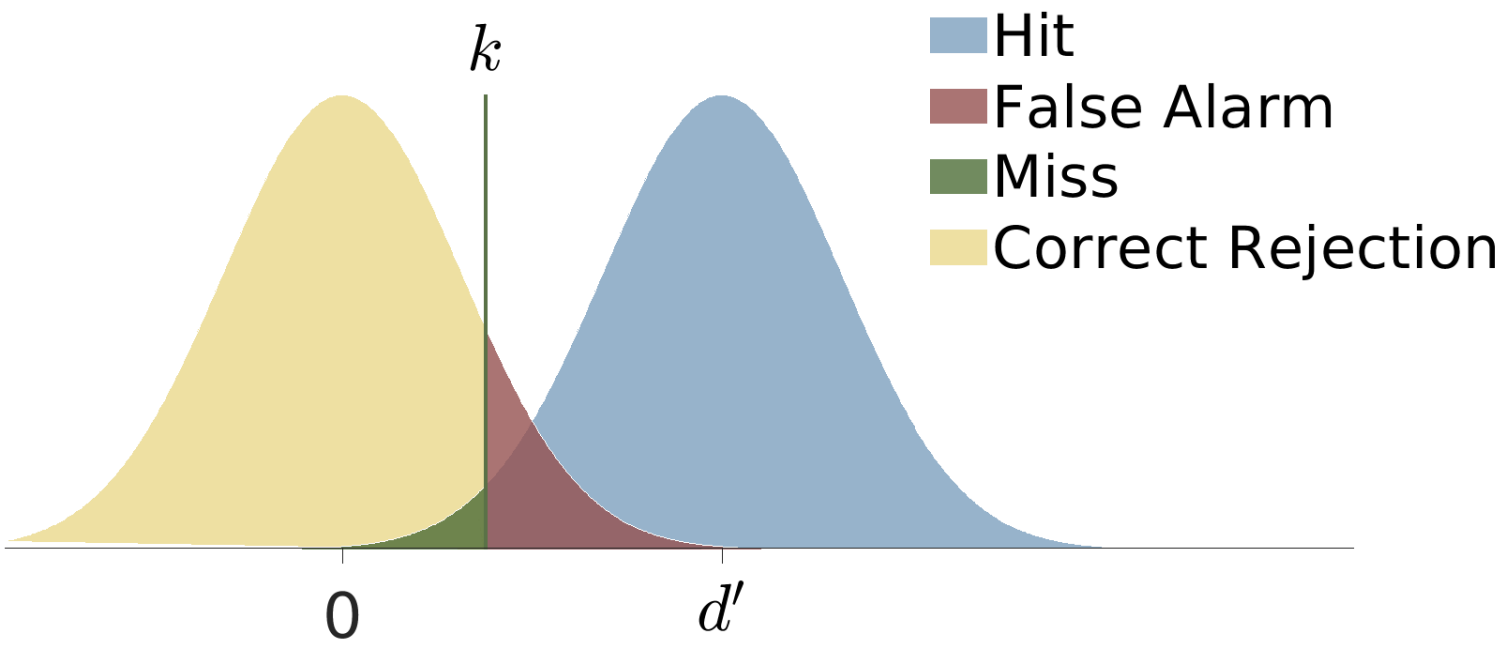
Signal Detection Model

- As a cognitive model SDT assumes there is a uni-dimensional latent continuum on which stimuli are experienced
  - experiences are sampled from signal and noise distributions with equal variances, and means separated by a discriminability d'
- The decision made depends on whether the momentary experience from the falls above or below a criterion k
  - incorporating the decision component goes beyond psychophysical models, like the Thurstone (1927) law of comparative judgment, in which responses are entirely determined by stimulus properties



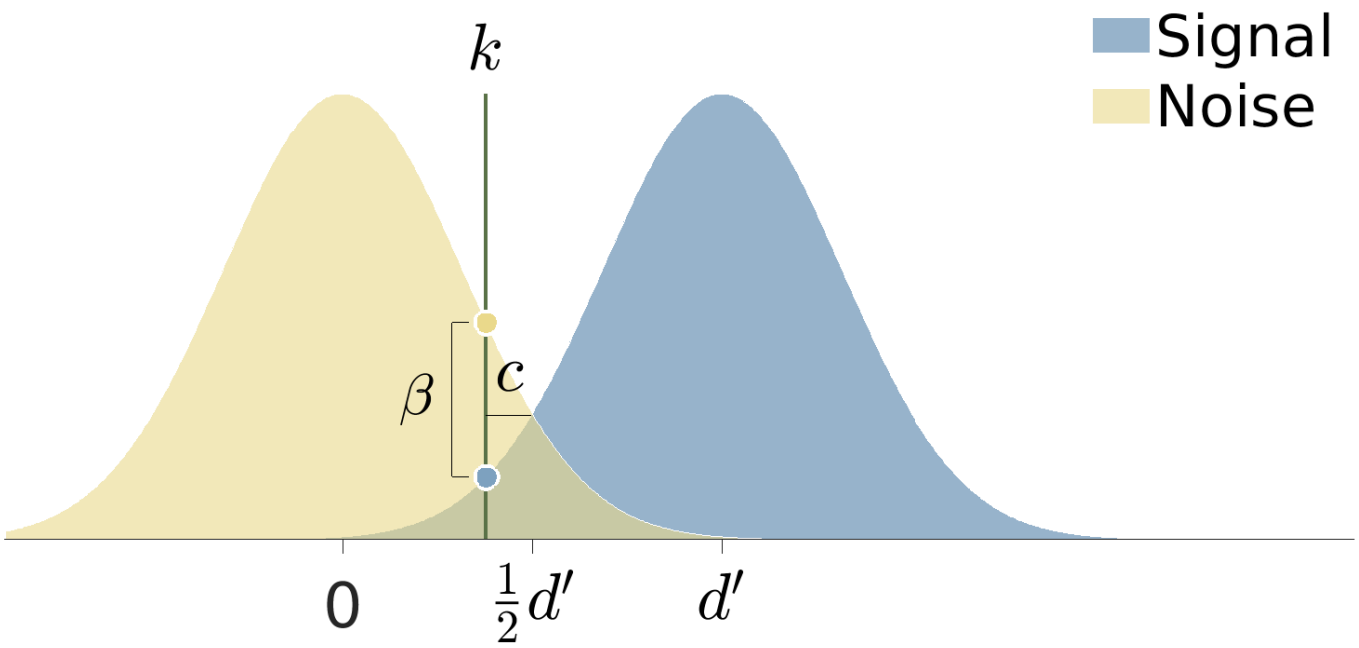
Signal Detection Model

- The sampling and decision process model defines probabilities that connect the
  - parameters: discriminability d', criterion k
  - derived parameters: hit, miss, false alarm and correct probabilities
  - data: hit, miss, false alarm and correct rejection counts



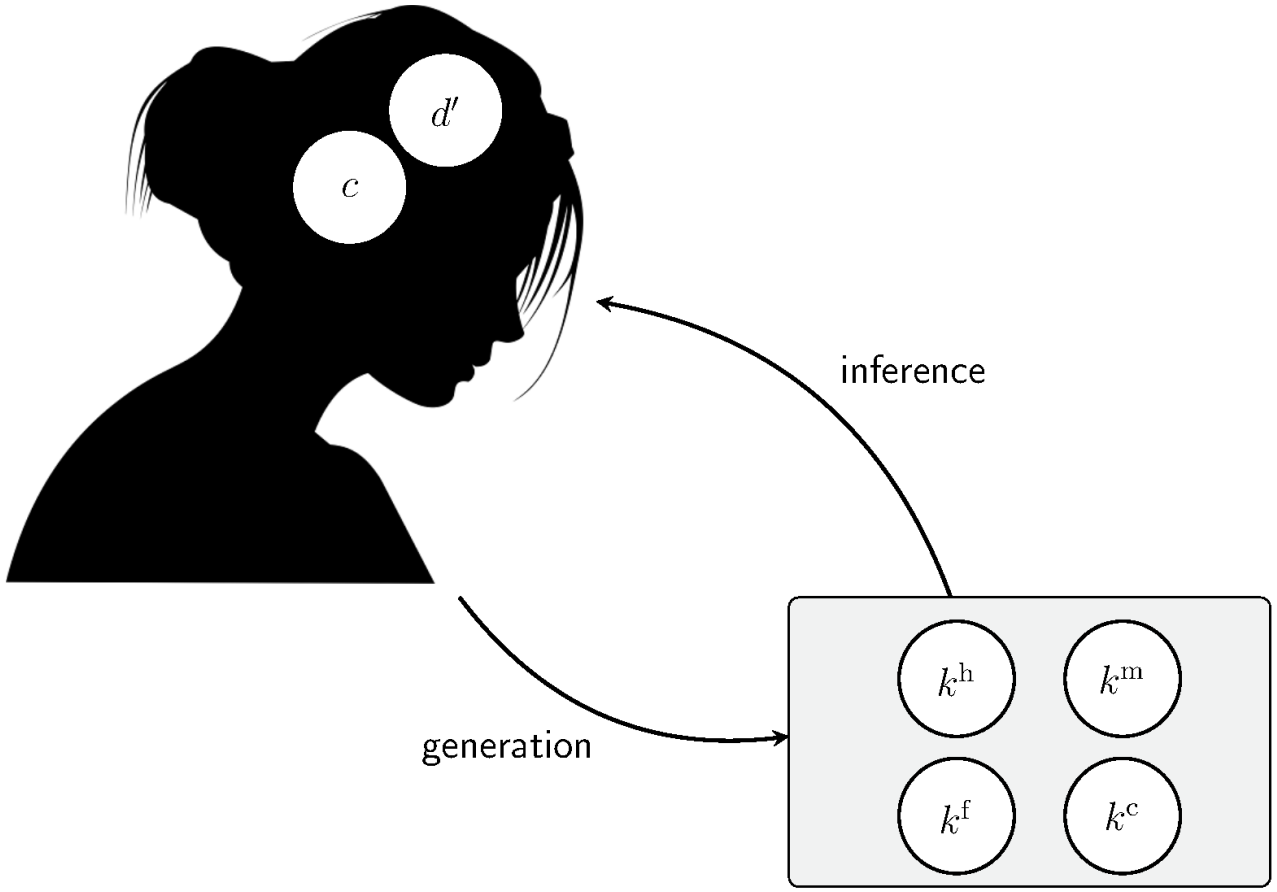
Signal Detection Model

- There are two alternative ways of measuring the bias of the criterion
  - the difference c between the criterion and the unbiased value $\frac{d'}{2}$
  - the likelihood ratio β of signal to noise



Two-Choice Signal Detection Model Structure

- The two-choice signal detection model has the structure
  - observed counts of hits k^{h}, false alarms k^{f}, misses k^{m}, and correct rejections k^{c}
  - latent parameters for discriminability d' and bias c (or k or β)



Wine Tasting

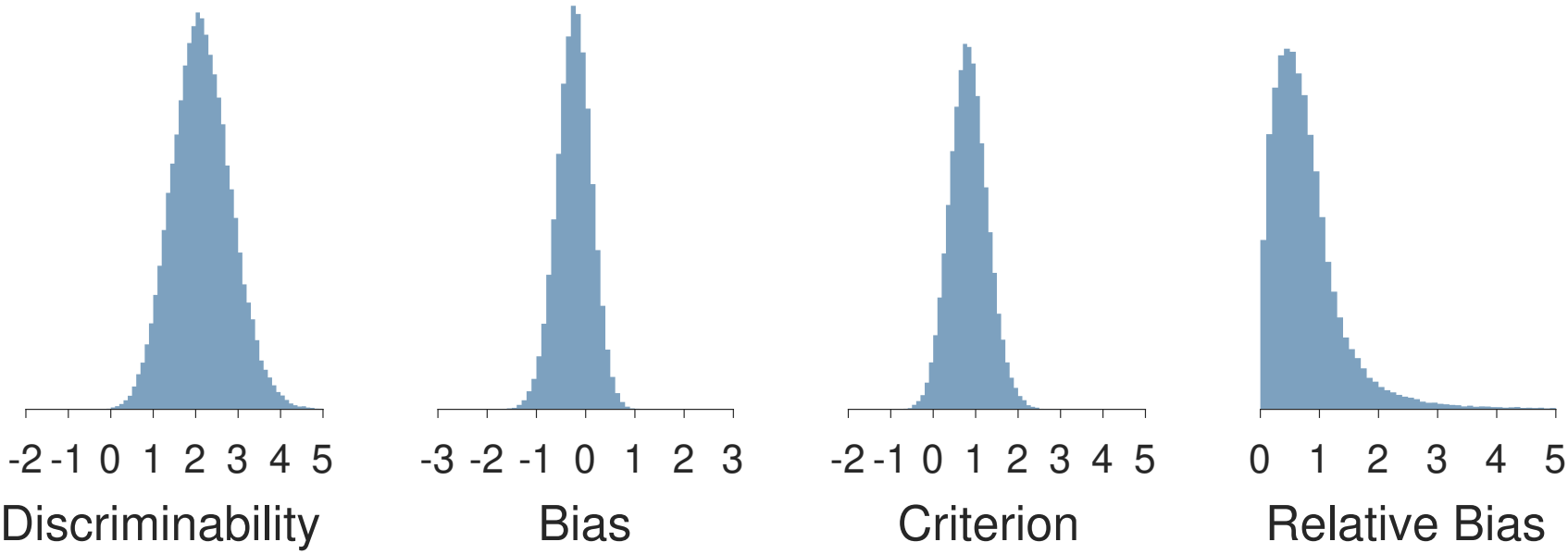
Wine Tasting Data (Abdi, 2007)

- There are 10 wines contaminated with Gamay and 10 non-contaminated ones
- The wine taster
  - detects Gamay for 9 of the 10 contaminated wines
    - corresponds to 9 hits and 1 miss
  - incorrectly identify Gamay in 2 of the 10 non-contaminated wines
    - corresponds to 2 false alarms and 8 correct rejections

	gamay	pure pinot noir
detect gamay	9	2
detect no gamay	1	8

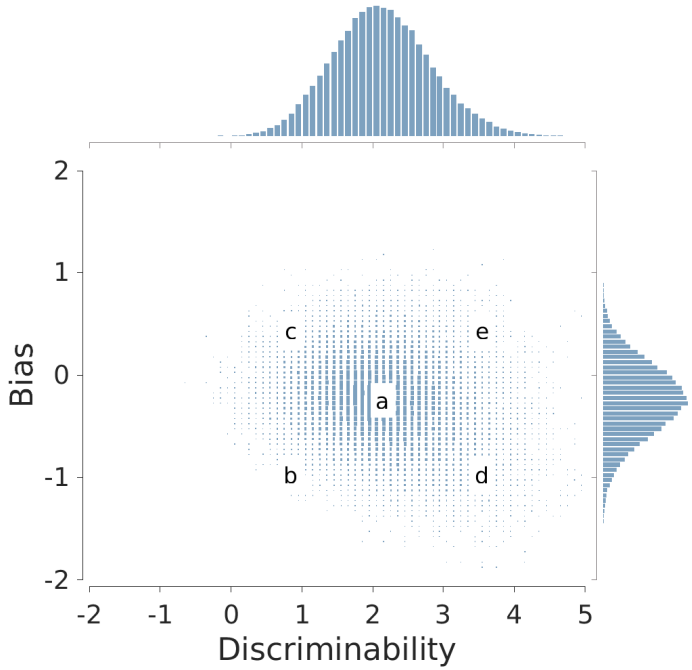
Marginal Posterior Distributions

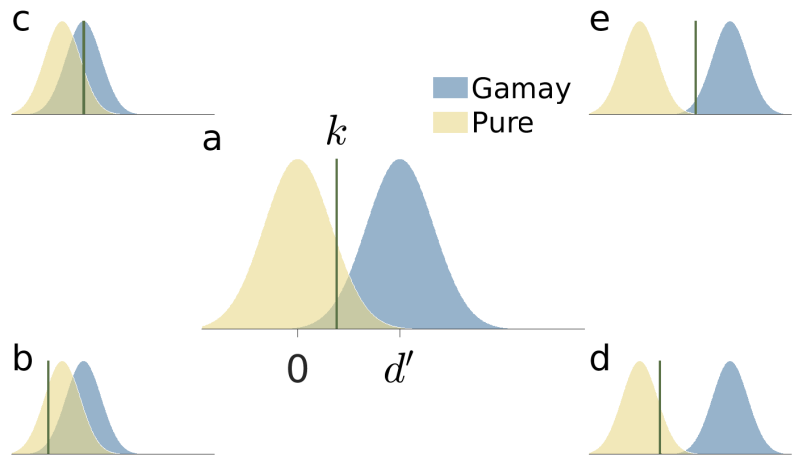
- The marginal posteriors show considerable uncertainty
  - mean posterior discriminability is $d' = 2.13$ and the 95% credible interval is $(0.81, 3.56)$
    - anywhere from modest to excellent detection ability is possible
  - mean posterior bias is $c = -0.25$ and the 95% credible interval is $(-0.95, 0.42)$
    - bias towards false alarms, or misses, or no bias, are all possible



Joint Distribution

- The joint distribution of discriminability d' and bias c is well approximated by the marginals
  - interpreting specific parameterizations with significant posterior density shows the range of possibilities

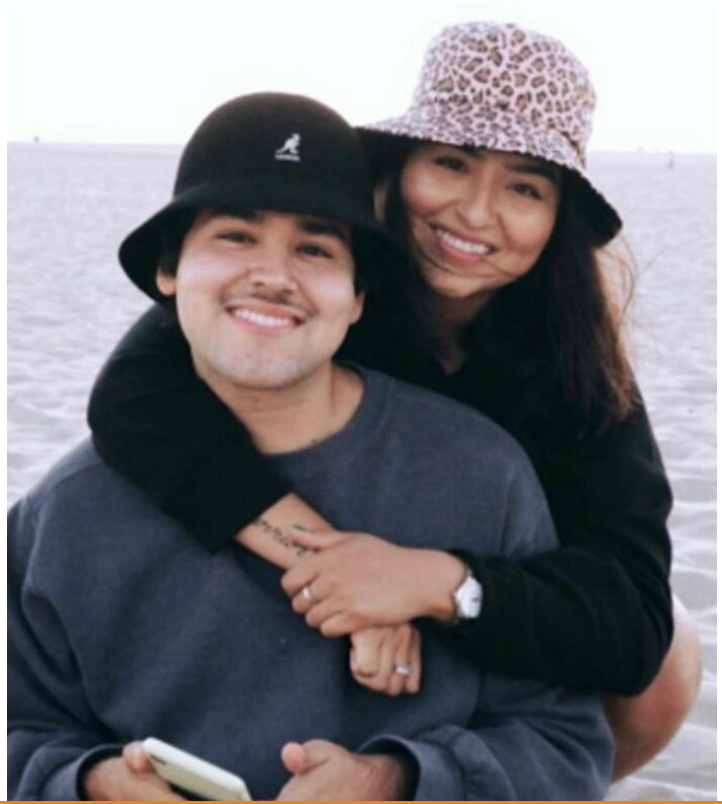


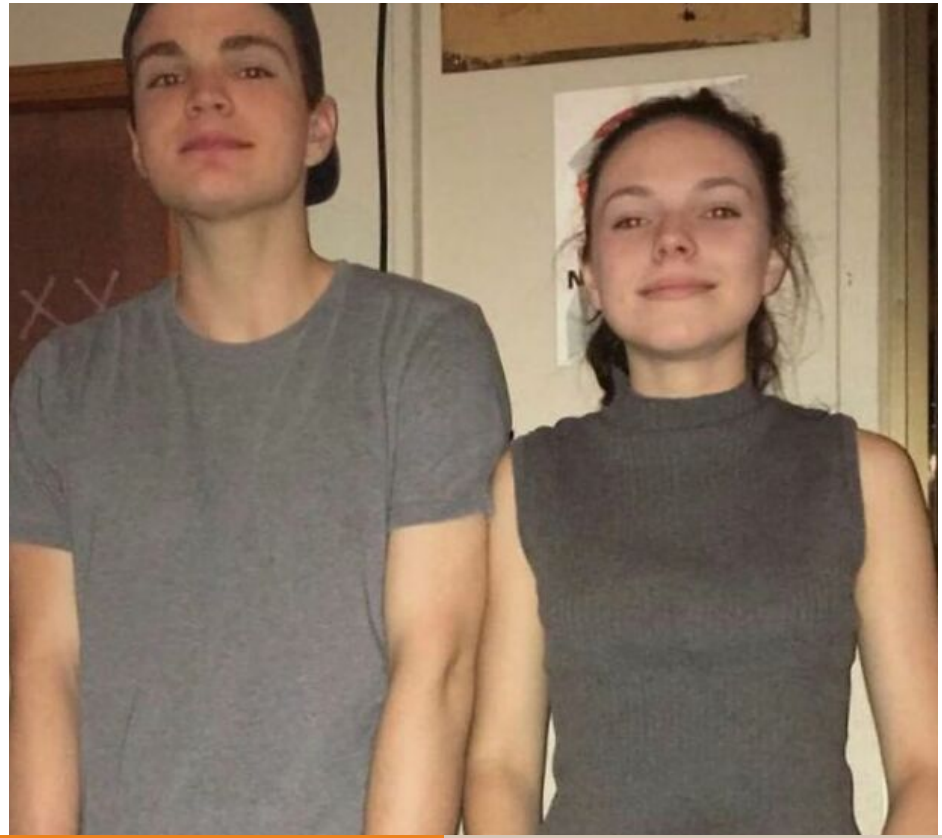


Siblings or Dating

Siblings or Dating

- An online challenge to decide whether a couple are siblings or dating based on a photo
  - used 127 images, 50 with "close contact" and 77 with "distant posing"
  - many of the questions are trick questions





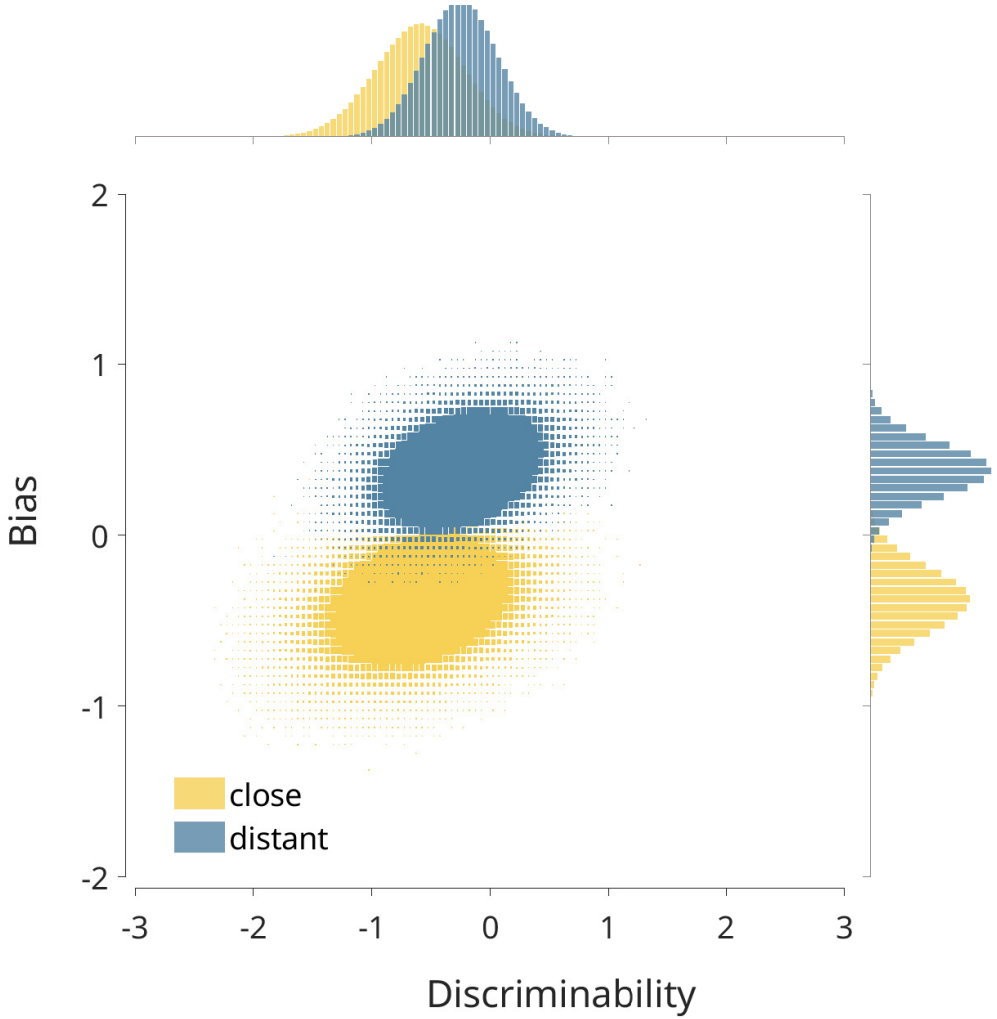
Siblings or Dating Data

- The data show more errors than correct decisions for both close and distant images
- We treat (arbitrarily) dating as signal and siblings as noise
  - few correct rejections for close images, and a tendency towards dating decisions
  - lots of misses for distant images, and a tendency towards sibling decisions overall

	truth			
	close contact		distant posing	
decision	dating	siblings	dating	siblings
dating	16	15	16	10
siblings	14	5	36	15

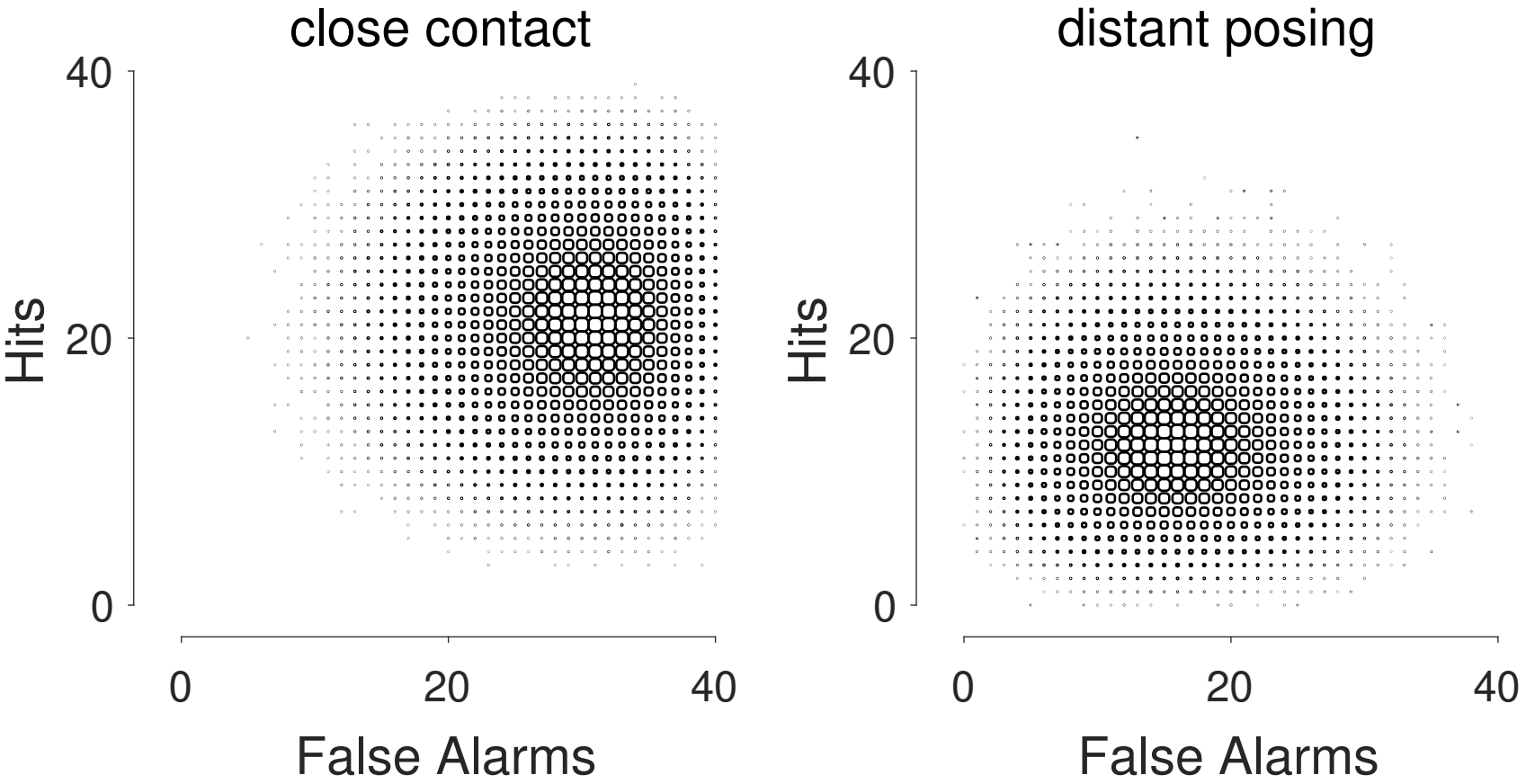
Joint and Marginal Posterior Distributions

- Discriminability is probably negative, reflecting the trick questions
  - bias probably favors dating decisions for close images and sibling decisions for distant images



Posterior Predictive Analysis

- Posterior predictive analysis is based on an imagined experiment with 40 dating (signal) and 40 siblings (noise) trials
  - provides a potentially more concrete way of understanding the model inferences



References i

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Macmillan, N. A. (2001). Signal detection theory. *International Encyclopedia of the Social & Behavioral Sciences*, 14075–14078.

Thurstone, L. L. (1927). A law of comparative judgement. *Psychological Review*, *34*, 273–286.